DALIAN NUMBER THEORY AND
ARITHMETIC GEOMETRY CONFERENCE

2026 大连数论与算术几何研讨会

会议手册

CONFERENCE HANDBOOK

大连理工大学
DALIAN UNIVERSITY OF TECHNOLOGY

復旦大學
FUDAN UNIVERSITY

会议日程

Conference Calendar | Aug 16–21, 2026

报到与入住：8月16日（周日）14:00-21:00 （当天 17:30-20:00 时段酒店一楼可点餐）
报到地点：大连理工国际会议中心酒店大堂
会议地点：大连理工大学综合教学 1 号楼 153 房间
早餐地点：酒店 1 楼；午餐、晚餐、晚宴地点：酒店 2 楼

	开幕日 · OPENING 08.17 星期一 MON	学术日 · DAY I 08.18 星期二 TUE	学术日 · DAY II 08.19 星期三 WED	学术日 · DAY III 08.20 星期四 THR	闭幕日 · CLOSING 08.21 星期五 FRI
09:20–09:30	**开幕式** **柳振鑫** *大连理工大学数学科学学院*				
09:30–10:30	**刘若川** *北京大学*	**扶磊** *清华大学*	**陈柯** *南京大学*	**肖梁** *北京大学*	**秦厚荣** *南京大学*
10:30–11:00	**茶歇及合影**	**茶歇**	**茶歇**	**茶歇**	**茶歇**
11:00–12:00	**范洋宇** *北京理工大学*	**丁一文** *北京大学*	**盛茂** *清华大学*	**许大昕** *中国科学院* *数学与系统科学研究院*	**欧阳毅** *中国科学技术大学*
12:00–14:30	**午餐**	**午餐**	**午餐**	**午餐**	**午餐**
14:30–15:30	**闵钰** *香港科技大学*	**胡永泉** *中国科学院* *数学与系统科学研究院*	**自由讨论** (14:30–17:30)	**胡昊宇** *南京大学*	**离会** (14:30–19:30)
15:30–16:00	**茶歇**	**茶歇**		**茶歇**	
16:00–17:00	**杜衡** *清华大学*	**王善文** *中国人民大学*		**赵和耳** *哈尔滨工业大学*	
17:00–17:30					
17:30–19:30	**晚餐**		**晚餐**	**晚餐**	
18:00–20:00		**会议晚宴**			

刘若川

北京大学

8月17日 9:30-10:30

报告题目：Some Examples of Eigencurves

摘要 · ABSTRACT

In joint work with Truong, Xiao and Zhao, we develop a theory on the geometry of eigencurves. In this talk, I will apply this framework to examine several concrete examples.

范洋宇

北京理工大学

8月17日 11:00-12:00

报告题目：Syntomic Formalism and p-adic Logarithms of Heegner Classes

摘要 · ABSTRACT

The Birch–Swinnerton-Dyer conjecture predicts a precise relation between the behavior of an L-function at its central point and the arithmetic of an elliptic curve. In the rank-one setting, Heegner classes provide a fundamental bridge between the analytic and arithmetic sides. I will explain a p-adic Waldspurger formula, valid also when p is non-split in the CM field, which relates the p-adic logarithms of generalized Heegner classes to special values of anticyclotomic p-adic L-functions. The main geometric input is syntomic formalism with coefficients, which allows the relevant Abel–Jacobi calculation to be carried out on semistable Shimura curves. Combined with anticyclotomic Iwasawa theory, this yields a p-converse theorem for self-dual CM characters, with applications to Sylvester’s conjecture on sums of two rational cubes and to Goldfeld’s conjecture for CM elliptic curves.

闵钰

香港科技大学

8月17日 14:30-15:30

报告题目：Congruences of Syntomic Cohomology

摘要 · ABSTRACT

Let K be a finite extension of Q_p and G_K be its absolute Galois group. Given two crystalline Z_p representations of G_K, we are interested in the relation between congruences of the two representations and congruences of their Bloch-Kato Selmer groups. But mod p^n reductions of representations do not contain enough information in general. In this talk, we will consider the mod p^n reductions of the prismatic F-gauges attached to these crystalline representations and explain how these reductions might determine the reductions of their first syntomic cohomology groups, which can be regarded as refinement of local Bloch-Kato Selmer groups.

杜衡

清华大学

8月17日 16:00-17:00

报告题目：p-adic Monodromy and Newton Polygons

摘要 · ABSTRACT

Berger’s theorem states that every de Rham Galois representation is potentially log-crystalline. I will discuss a relative version of this theorem, conjectured by Liu and Zhu, for de Rham local systems. The main new ingredient is the Newton polygon function attached to a de Rham local system. I will show that a relative p-adic monodromy theorem holds over Newton-constant loci. In particular, the Liu–Zhu conjecture holds on a dense open subset. Near rank-one adic points, our criterion is also sharp. I will also explain an application to a recent conjecture of Howe and Klevdal concerning potential good reduction for admissible pairs.

扶磊

清华大学

8月18日 9:30-10:30

报告题目：Arithmetic Hypergeometric D-modules and Exponential Sums for Reductive Groups

摘要 · ABSTRACT

For a family of representations of a reductive group, we define a Laurent polynomial on the group. The exponential sum associated to this Laurent polynomial is called the hypergeometric exponential sum. We introduce an arithmetic hypergeometric D-module to study the exponential sum. We show it is overholonomic, determine the open set where it is an overconvergent F-isocrystal, and estimate its rank. We apply these results to the estimation of the exponential sum. This is a joint work with Xuanyou Li and Chenhan Liu.

丁一文

北京大学

8月18日 11:00-12:00

报告题目：Classicality of Weight One Modular Forms

摘要 · ABSTRACT

We give a new proof of the classicality of weight one modular forms. This is a joint work with Tian Qiu and Zhixiang Wu.

胡永泉

中国科学院数学与系统科学研究院

8月18日 14:30-15:30

Multivariable (phi, Gamma)-modules and the locality question for GL_2

摘要 · ABSTRACT

Let K be a finite unramified extension of Q_p and let rho be a 2-dimensional mod p Galois representation of G_K. In the mod p Langlands program we aim to associate to rho a smooth mod p representation pi(rho) of GL_2(K). If rho arises globally from automorphic forms, there exists a non-canonical global candidate for pi(rho). In this talk, we will survey known results concerning pi(rho) and its attached multivariable (phi,Gamma)-modules, and address the locality question. This is joint work with Breuil, Herzig, Koziol, Morra, Schraen and Shin.

王善文

中国人民大学

8月18日 16:00-17:00

报告题目：p-adic Hahn Series with Sparse Support

摘要 · ABSTRACT

Let p be a prime number. We introduce a sparseness condition on the supports of p-adic Hahn series, and prove that this condition implies transcendence over the completion of maximal unramified extension of Q_p. As an application, we prove our order type conjecture for algebraic p-adic Hahn series with bounded support under the condition that the support has only finitely many accumulation points. This is joint work with Yijun Yuan.

陈柯

南京大学

8月19日 9:30-10:30

报告题目：On Images of Galois Representations Associated to One-motives

摘要 · ABSTRACT

Galois representations associated to abelian varieties over a field in characteristic zero are expected to have nice properties, including variants of open image in suitable l-adic or adelic Lie groups. We discuss similar properties for one-motives and related results.

盛茂

清华大学

8月19日 11:00-12:00

报告题目：Nonlinear Hodge Correspondence in Positive Characteristic

摘要 · ABSTRACT

I shall report on my recent work on nonlinear Hodge correspondence in positive characteristic. It is a correspondence between certain category of nonlinear connections with nilpotent p-curvatures and certain category of nonlinear Higgs fields with nilpotency condition. I shall also introduce the notion of a nonlinear Fontaine module, and explain that the relative nilpotent de Rham moduli, equipped with the nonabelian Gauss-Manin connection, the nonabelian Hodge filtration and the nonabelian Frobenius structure induced by the Ogus-Vologodsky correspondence, is naturally a nonlinear Fontaine module.

肖梁

北京大学

8月20日 9:30-10:30

报告题目：TBD

摘要 · ABSTRACT

TBD

许大昕

中国科学院数学与系统科学研究院

8月20日 11:00-12:00

报告题目：Comparison between Algebraic and Arithmetic D-modules

摘要 · ABSTRACT

Berthelot’s theory of arithmetic D-modules provides a p-adic analogue of the sheaf-theoretic formalism on varieties of characteristic p. In this talk I will give a brief review of this theory. A central point will be the comparison between algebraic D-modules on the generic fiber and arithmetic D-modules on the special fiber. This comparison is the mechanism that allows one to put Frobenius structures on rigid algebraic connections.

胡昊宇

南京大学

8月20日 14:30-15:30

报告题目：A Generalization of Deligne's Finiteness Theorem

摘要 · ABSTRACT

In 2012, Esnault and Kerz proved the Deligne’s finiteness theorem for l-adic sheaves, which says that the number of geometrically irreducible l-adic local systems on a smooth variety over a finite field, with bound ramification along a normal compactification and ranks, is finite. In this lecture, I will present a generalization and a new proof of this theorem. The new ingredient is a universal bound of Betti numbers for étale sheaves with wild ramifications. This is a joint work with Jean-Baptiste Teyssier.

赵和耳

哈尔滨工业大学

8月20日 16:00-17:00

报告题目：Reduction Types of Rigid 1-motives via Galois Representation

摘要 · ABSTRACT

We introduce rigid 1-motives, formal 1-motives, log formal 1-motives, study relations among them, define reduction types of rigid 1-motives, as well as l-adic realizations (l can be p) of various 1-motives. In the end, we give a Neron-Ogg-Shafarevich criterion of the reduction types of rigid 1-motives via l-adic realizations (l can be p). This is base on a part of a joint work with Xu Shen and Khai-Hoan Nguyen-Dang.

秦厚荣

南京大学

8月21日 9:30-10:30

报告题目：The CM Conductor and the Lang-Trotter Conjecture for CM Elliptic Curves

摘要 · ABSTRACT

Whether a quadratic polynomial can represent primes infinitely often - for example, Euler’s conjecture that x^2+1 can represent primes infinitely often - is notoriously difficult and remains wide open. More generally, the Hardy - Littlewood conjecture gives an asymptotic formula for the count of such primes. In parallel, for an elliptic curve E over Q with complex multiplication (CM) and a fixed nonzero integer r, the Lang-Trotter conjecture predicts an asymptotic formula for πE,r(x) = #{p ⩽ x : a_p = r}, where a_p is the Frobenius trace. We introduce a new invariant, the CM conductor K_E, which refines the Serre conductor in the CM case. Using K_E, we give a corrected formula for the Lang-Trotter constant c_E,r, differing from the previous one by Baier and Jones. Assuming the Hardy-Littlewood conjecture on quadratic polynomial primes, we prove that our version of the Lang-Trotter conjecture holds for every CM elliptic curve over Q. We compute the constants explicitly for each imaginary quadratic field of class number one, characterize exactly when the constant vanishes, and prove the conjecture unconditionally in the vanishing case. Numerical evidence strongly supports our formulation. This is joint work with Longxi Hu and Kaisheng Lei.

欧阳毅

中国科学技术大学

8月21日 11:00-12:00

报告题目：Isogeny-Based Cryptography

摘要 · ABSTRACT

本报告将介绍基于超奇异椭圆曲线和阿贝尔簇同源的后量子密码体系，并介绍我们团队在这方面的一些工作，主要包括后量子密码签名方案 SQIsign2D^2 的设计。

交通指南

> 会议提供机场接送机服务，如您不需要接送机服务，欢迎参考以下交通方式前往会议地点。

✈ 大连周水子国际机场

距离：约 13 公里

出租车/网约车：

- 约25 – 30分钟，35 – 50元

公共交通：

- 全程约60分钟，票价约5 – 7元

🚆 大连站

距离：约13公里

出租车/网约车：

- 约25 – 30分钟，35 – 45元

公共交通：

- 全程约50 – 60分钟，票价约5 – 7元

🚄 大连北站

距离：约 20 公里

出租车/网约车：

- 约30 – 40分钟，50 – 60元

公共交通：

- 全程约60 – 70分钟，票价约5 – 7元

校园地图

大连理工大学 凌水主校区

大连理工大学数学科学学院简介

大连理工大学原应用数学系创建于1949年；1958年全国性院系调整时被取消，成立应用数学专业，隶属数理力学系；1979年应用数学系得以恢复重建，设应用数学本科专业，同年开始招收硕士研究生。1986年计算数学专业获得博士学位授予权；1988年被批准设立数学博士后流动站；1998年运筹学与控制论专业获得博士学位授予权；2002年计算数学博士点被评为国家重点学科；2003年获数学一级学科博士学位授予权，涵盖基础数学、计算数学、概率论与数理统计、应用数学和运筹学与控制论5个二级学科博士点，并经教育部批准自主增设金融数学与保险精算博士点；2008年被批准为辽宁省重点一级学科，同年被批准为“国家理科基础科学研究和教学人才培养基地”。2009年，数学科学学院成立。2012年与中国科学院数学与系统科学研究院联合设立华罗庚班；2019年，数学与应用数学专业获批教育部首批“双万计划”国家一流本科专业建设点，“计算数学与数据智能重点实验室”获批辽宁省重点实验室；2020年，数学与应用数学专业入选教育部“强基计划”，信息与计算科学专业获批教育部“双万计划”国家一流本科专业建设点，辽宁省科技厅批准大连理工大学成立“辽宁应用数学中心”；2021年，“华罗庚数学拔尖学生培养基地”获批教育部“基础学科拔尖学生培养计划2.0基地”；2023年，入选本科教育教学改革试点工作计划（101计划）。

经过几代人不懈努力，特别是近年来通过“211工程”、“985工程”和“双一流”建设的实施，学科的学术水平和综合能力得到明显提高。ESI国际学科排名前1%,QS世界排名350位左右。学院近5年承担国家自然科学基金项目和省部级科研项目100余项，在数学若干重要前沿领域取得具有国际水平的成果。

数学科学学院现有5个研究所：数学研究所、计算科学研究所、统计与金融研究所、应用数学研究所、运筹学与控制论研究所；3个中心：辽宁应用数学中心、数学教学中心和数学研究中心；2个实验室：辽宁省计算数学与数据智能重点实验室、计算几何与图形图像实验室。主办核心学术期刊《数学研究及应用》（英文版）。数学科学学院现有专任教师97人，其中教授44人、副教授48人、讲师2人、助理教授3人、正编审1人，其中国务院学科评议组成员1人、教育部教学指导委员会委员2人、国家教学名师2人、辽宁省数学类专业教指委主任委员1人、辽宁省教学名师5人、宝钢优秀教师奖获得者7人；国家杰出青年科学基金获得者1人、国家级青年人才10人、教育部跨世纪人才1人、教育部新世纪人才4人。

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